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Subject: Programming with JAVA	Sem: III AY: 2026-27
Assignment No: 16	
TITLE: Write a Java program to create and handle a user-defined exception.	

Problem Statement

Write a Java program to create and handle a user-defined exception.

Learning Objectives

To create and implement custom exceptions for application-specific error handling.

Theory:

User-defined exceptions are created by extending the Exception class. They allow developers to

define meaningful exceptions for specific business logic. Custom exceptions improve program

readability and simplify debugging by providing descriptive error messages. They are widely used in enterprise software and validation systems.

Exercise

1. Create a Voting System that throws a custom exception if age is below 18

```
import java.util.Scanner;
```

```
class InvalidAgeException extends Exception {  
    public InvalidAgeException(String message) {
```

```
        super(message);  
    }  
}
```

```
public class VotingSystem {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter your age: ");  
        int age = sc.nextInt();  
  
        try {  
            if (age < 18) {  
                throw new InvalidAgeException("You are not eligible for voting.");  
            }  
  
            System.out.println("You are eligible for voting.");  
  
        } catch (InvalidAgeException e) {  
            System.out.println("Exception: " + e.getMessage());  
        }  
  
        sc.close();  
    }  
}
```

```
}
```

```
Enter your age: 21
You are eligible for voting.
PS C:\Users\Arpit\Java_assignment_2> ^C
PS C:\Users\Arpit\Java_assignment_2>
PS C:\Users\Arpit\Java_assignment_2> c:: cd 'c:\Users\Arpit\Java_assignment_2'; & 'C:\Program Files\Java\jdk-26.0.1\
bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Arpit\AppData\Roaming\Code
\User\workspaceStorage\9e128c6022f184db52880e45cf51c1e4\redhat.java\jdt_ws\Java_assignment_2_f33b7c19\bin' 'VotingSys
tem'
Enter your age: 16
Exception: You are not eligible for voting.
PS C:\Users\Arpit\Java_assignment_2> █
```

2. Create a Driving License System that throws a custom exception if the user's age is below 18. If the age is valid, display that the user is eligible for a driving license

```
import java.util.Scanner;
```

```
class InvalidAgeException extends Exception {

    public InvalidAgeException(String message) {

        super(message);

    }

}
```

```
public class DrivingLicenseSystem {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");

        int age = sc.nextInt();

        try {

            if (age < 18) {
```

```

        throw new InvalidAgeException("You are not eligible for a driving license.");
    }

    System.out.println("You are eligible for a driving license.");

} catch (InvalidAgeException e) {

    System.out.println("Exception: " + e.getMessage());

}

sc.close();

}

}

```

OUTPUT:

```

Enter your age: 21
You are eligible for a driving license.
PS C:\Users\Arpit\Java_assignment_2> ^C
PS C:\Users\Arpit\Java_assignment_2>
PS C:\Users\Arpit\Java_assignment_2> c:: cd 'c:\Users\Arpit\Java_assignment_2'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-p
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Arpit\AppData\Roaming\Code\User\workspaceStorage\9e128c6022f184db52880e45cf5
1c1e4\redhat.java\jdt_ws\Java_assignment_2_f33b7c19\bin' 'DrivingLicenseSystem'
Enter your age: 17
Exception: You are not eligible for a driving license.
PS C:\Users\Arpit\Java_assignment_2> █

```

Conclusion

We learned to create and handle user-defined exceptions in Java. The programs successfully validate age and display appropriate messages for voting and driving license eligibility.